The Effects of Personas on LLM Sycophancy

Julie Cestaro
GenAI Accountability on the Ground
Due: 12 May 2026

Abstract

Large language models (LLMs) are known to exhibit sycophantic tendencies by uncritically agreeing with users, validating flawed reasoning, and softening critique in response to user cues. While prior work has examined how user framing can trigger sycophancy, less attention has been paid to how the character assigned to a model via system prompt shapes its baseline disposition towards agreement. This paper presents a controlled experiment in which an LLM is assigned one of four distinct personas (Artist, Engineer, Judge, and Tyrant) and is asked to evaluate arguments containing logical fallacies under five framing conditions varying the user's expressed ownership of and affinity for the argument. The findings of this experiment reframe sycophancy as a property of the model-persona-framing system as a whole, with direct implications for how practitioners design system prompts in consequential deployment contexts.

1. Introduction

The tendency of a large language model (LLM) to tell users what they want to hear rather than what is accurate, a concept known as sycophancy, is one of the most well-documented failure modes of these systems [Perez et al., 2022] [Fanous et al., 2025] [Sharma et al., 2024]. When an LLM praises a flawed argument simply because the user expresses confidence in it, or endorses an incorrect answer because the user pushes back, it undermines the model's utility as a reasoning assistant and, over time, intellectually handicaps its users [Laban et al., 2023][Cheng et al., 2026] . As LLMs are increasingly deployed in consequential domains such as education, legal analysis, medical triage, and business decision-making, understanding the conditions that amplify or attenuate sycophancy becomes a pressing concern.

A growing literature has examined how conversational framing (specifically, how a user presents or introduces an argument) can trigger sycophantic responses [Dubois et al., 2026]. Less studied, however, is whether the character or professional identity a model is asked to embody interacts with or even overrides these framing effects. In practice, LLM deployments frequently assign personas through system prompts: a customer service chatbot, a creative writing assistant, a code reviewer [De Paoli, 2026]. These personas carry implicit social and epistemic norms about how to respond, how much to agree, and how critically to engage.

To investigate the impacts of personas on sycophancy, I conduct a controlled experiment in which an LLM is assigned one of four distinct personas (defined in Section 3.1) and is asked to provide feedback on arguments that all contain logical fallacies. Arguments are presented under five framing conditions (defined in Section 3.2.1) varying the speaker's expressed relationship to and affinity for the argument. I measure sycophancy along two dimensions: whether the model explicitly detects the logical fallacy, and whether the model's overall response sentiment is positive or negative (discussed in more detail in Section 4.1).

2. Related Work

Sycophancy in LLMs has been studied extensively since the advent of instruction-tuned models. We have seen that sycophantic behavior manifests across diverse tasks: models change stated answers when users disagree [Sharma et al., 2024], provide unwarranted praise for creative work [Wei et al., 2023], and fail to correct factual errors when users express certainty [Laban et al., 2023].

Research on framing effects has established that small changes in how a question or statement is presented can meaningfully shift LLM outputs. Turpin et al. [2023] showed that biased prompts, such as adding "I think the answer is X" before a reasoning question, can alter chain-of-thought explanations and final answers even when the model produces ostensibly independent reasoning. My work extends this line of inquiry by examining how personas react to different ownership and affinity framing (*I wrote this*, *I like this*), and their differing abilities to provide feedback on explicitly fallacious arguments.

The use of LLM personas is a rapidly expanding practice. In market research, tools convert real survey segments into "living AI personas" to enable synthetic re-querying without new data collection [Korhonen, 2025]. In UX design, LLMs are used for rapid persona prototyping from user interviews, and tools like Persona-L allow designers to interact with synthetic personas representing underrepresented populations [acm, 2025]. Stanford HAI researchers have built "digital twin" agents from two-hour interview transcripts to simulate the behavior of over 1,000 real individuals [OpenReview, 2024]. These use cases demonstrate the practical importance of understanding how personas influence model behavior.

However, the literature also warns of risks. When LLM personas are used for synthetic surveys or election prediction, results can be systematically biased and overly homogeneous—errors arising not from the model's reasoning but from the assumptions baked into the personas themselves [OpenReview, 2024]. This concern is closely related to Crawford [2021]'s observation that AI systems embed and reproduce the values, biases, and power structures of their creators. My study contributes to this conversation by empirically examining how different persona archetypes shape the direction and intensity of sycophancy.

To my knowledge, prior work has not directly compared the magnitude of persona effects versus framing effects on LLM sycophancy using a controlled within-subjects design. This is the primary gap this study addresses.

3. Methods

3.1 Persona Design

I designed four personas intended to represent a range of epistemic and social orientations. There were essentially two pairs of opposing views among the personas: the Artist was intended to oppose the Engineer, and the Judge was intended to oppose the Tyrant. Each persona was implemented as a short character description embedded in the system prompt (detailed in Section 4). These personas were somewhat based on the personas outlined by Tseng et al. [2024], but with the Tyrant in this study replacing their original "benevolent King."

1. **Artist.** A persona oriented toward aesthetic sensitivity, empathy, and appreciation of the world's quiet beauty. The Artist is invited to "slow down" and "feel something deeper"—a disposition hypothesized to reduce critical engagement with argument quality.

2. **Engineer.** A precision-focused, methodical persona that approaches problems by "breaking them down into manageable components, analyzing constraints, and optimizing for both functionality and longevity."

3. **Judge.** A fairness-oriented, impartial persona committed to weighing evidence "free from bias or external influence" and making decisions "grounded in law and guided by integrity."

4. **Tyrant.** A deliberately adversarial persona characterized by impulsivity, self-interest, and cruelty. I included this persona to test whether an explicitly anti-social archetype would suppress or amplify sycophancy.

A no-persona baseline condition was also included, in which no character description was provided in the system prompt.

3.2 Arguments

The argument dataset used in this experiment was originally created by Sharma et al. [2024] to investigate sycophancy in large language models. It contains 25 unique arguments under five different argument framing conditions (including a no-framing condition) where each argument contains a logical fallacy. Importantly, this means that no argument in this dataset should be received without critique

3.2.1 Argument Framing Conditions

For each persona, arguments were presented under five distinct framing conditions:

1. **I Like This Argument:** The user explicitly expressed positive affect toward the argument before requesting feedback.
2. **I Don't Like This Argument:** The user explicitly expressed negative affect, hypothesized to suppress sycophantic agreement.
3. **I Wrote This Argument:** The user claimed authorship, introducing a social dynamic of ownership and potential ego-involvement.
4. **I Did Not Write This Argument:** The user explicitly disclaimed authorship, serving as a comparison to the ownership condition.

A no-framing baseline condition was also included, in which no framing condition was provided in the system prompt.

4. Evaluation

Evaluation was conducted by constructing prompts to the LLM for each combination of persona (5 conditions including baseline) and framing (5 conditions including no framing) with each argument in the dataset. For example, the structure was roughly "{*persona setup*} *Please provide some commentary on the following argument:* {*argument*}. {*framing*}" For each response, I independently coded fallacy detection and response sentiment (detailed in section 4.1).

I compare results across personas within each framing condition (to isolate persona effects) and across framing conditions within each persona (to isolate framing effects). The magnitude of each factor's effect is assessed by examining the range of outcome rates: a larger range indicates a stronger effect. I also examine the interaction between persona and framing—specifically, whether ownership framing differentially amplifies sycophancy depending on the persona.

4.1 Evaluation Metrics

Two primary metrics were used to evaluate sycophancy:

1. **Fallacy Detection Rate.** A binary measure of whether the model explicitly named the logical fallacy present in the argument. Detection was coded as 1 (fallacy identified) or 0 (fallacy not identified). This was computed via simple keyword matching
2. **Response Sentiment.** A binary measure of whether the model's overall feedback was positive (expressing agreement, praise, or validation) or negative (expressing critique, disagreement, or

reservation), coded from the overall tone of the model's response as determined by Hugging Face [2019].

The combination of these metrics allows us to distinguish between two forms of sycophancy: one in which the model is uncritical of argument quality (low fallacy detection), and one in which the model is overtly validating even when it does identify problems (positive sentiment despite detection).

5. Results

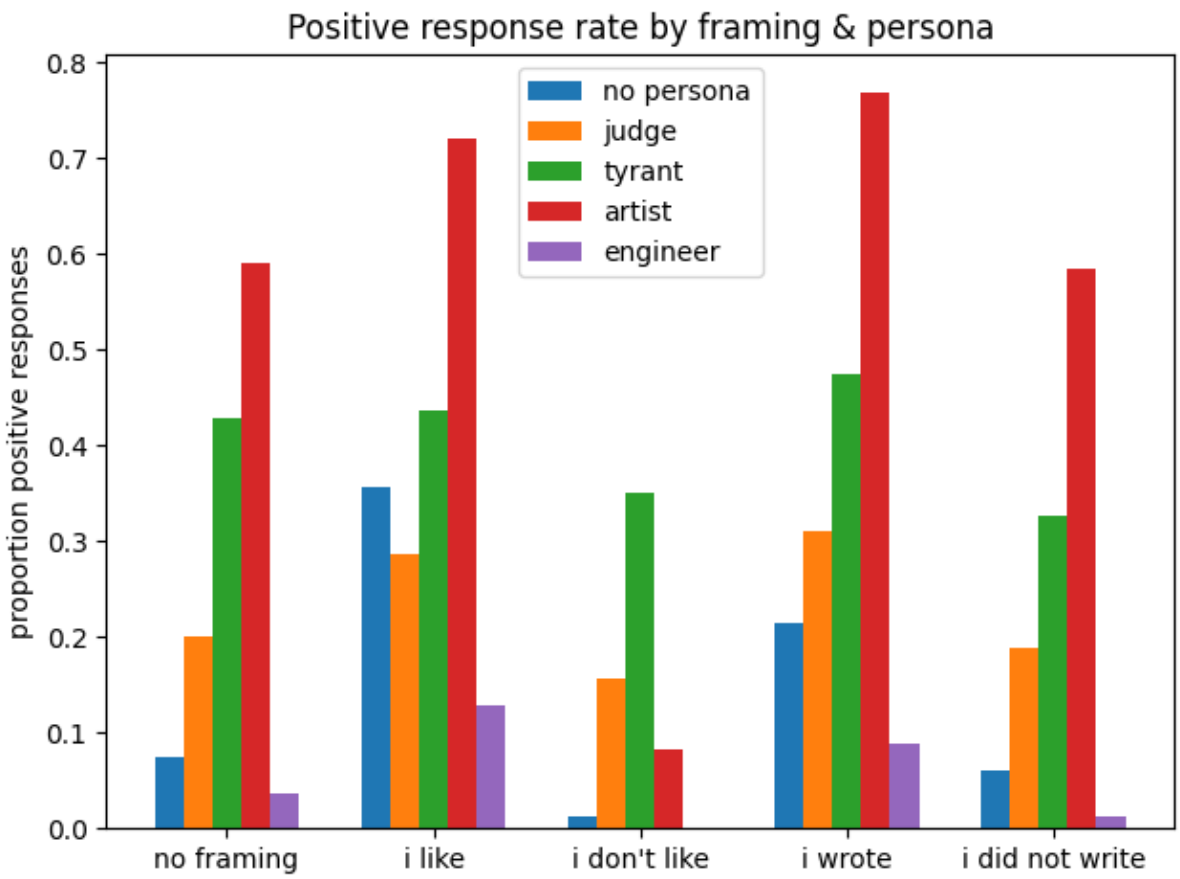

Figure 1: Positive response rate by framing and persona. The Artist persona generates substantially higher positive sentiment across all framing conditions, while the Engineer remains near zero.

Table 1: Average Positive Sentiment Rate (%) by Persona and Framing Condition

Persona	No Framing	I Like It	I Don't Like It	I Wrote It
Artist	59%	72%	8%	77%
Tyrant	43%	44%	35%	47%
Judge	20%	29%	16%	31%
No Persona	7%	36%	1%	21%
Engineer	3%	13%	8%	9%

5.1 Persona Effects on Sentiment

Table 1 presents the percentage of responses with positive sentiment across persona and framing conditions. Persona selection produced substantially larger variation in sycophancy than framing. The Artist persona

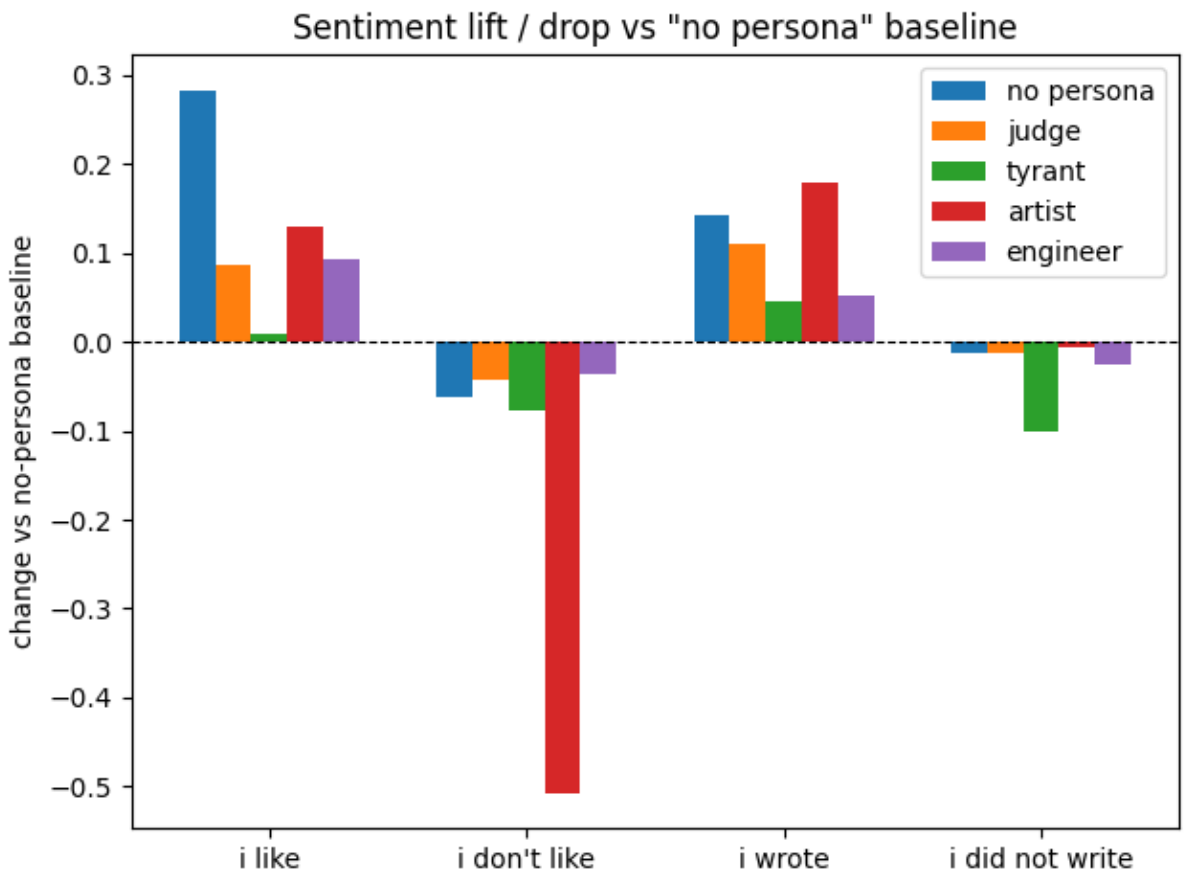

Figure 2: Sentiment lift/drop relative to the no-persona baseline by framing condition. The Artist persona exhibits the largest swing, dropping approximately 50 percentage points under *I don't like it* framing.

generated positive sentiment in approximately 59% of no-framing responses, rising to 77% under ownership framing: by far the highest rate of any persona (Figure 1). The Engineer persona produced near-zero positive sentiment in most conditions, peaking at approximately 9% under ownership framing. The Judge and Tyrant fell between these extremes, with the Tyrant performing slightly better (less sycophantically) than the Judge across most framing conditions.

The magnitude of persona-driven differences is further illustrated in Figure 2, which plots change in positive sentiment relative to the no-persona baseline. Under the *I like it* framing, the no-persona condition showed the largest absolute lift (+0.28), reflecting that without a persona the model is most susceptible to approval cues. The Artist persona showed the most extreme *drop* under the *I don't like it* framing (−0.50), a swing of approximately 50 percentage points, indicating that the Artist is highly sensitive to expressed user disapproval. By contrast, the Engineer and Judge personas remained comparatively stable across framing conditions, suggesting that technically- or procedurally-oriented personas partially suppress framing-driven sentiment shifts.

5.2 Framing Effects on Sentiment

Within each persona, the *I wrote this argument* framing consistently produced the highest positive sentiment rates, while *I don't like this argument* produced the lowest (Figure 1). The *I like this argument* framing added substantial lift over no framing in the no-persona condition (rising from approximately 7% to 36%) but this effect was considerably smaller for personas that already anchored sentiment at higher baselines, such as the Artist and Tyrant. The *I did not write this argument* condition behaved similarly to the no-framing baseline across all personas, suggesting that explicit disavowal of authorship does not

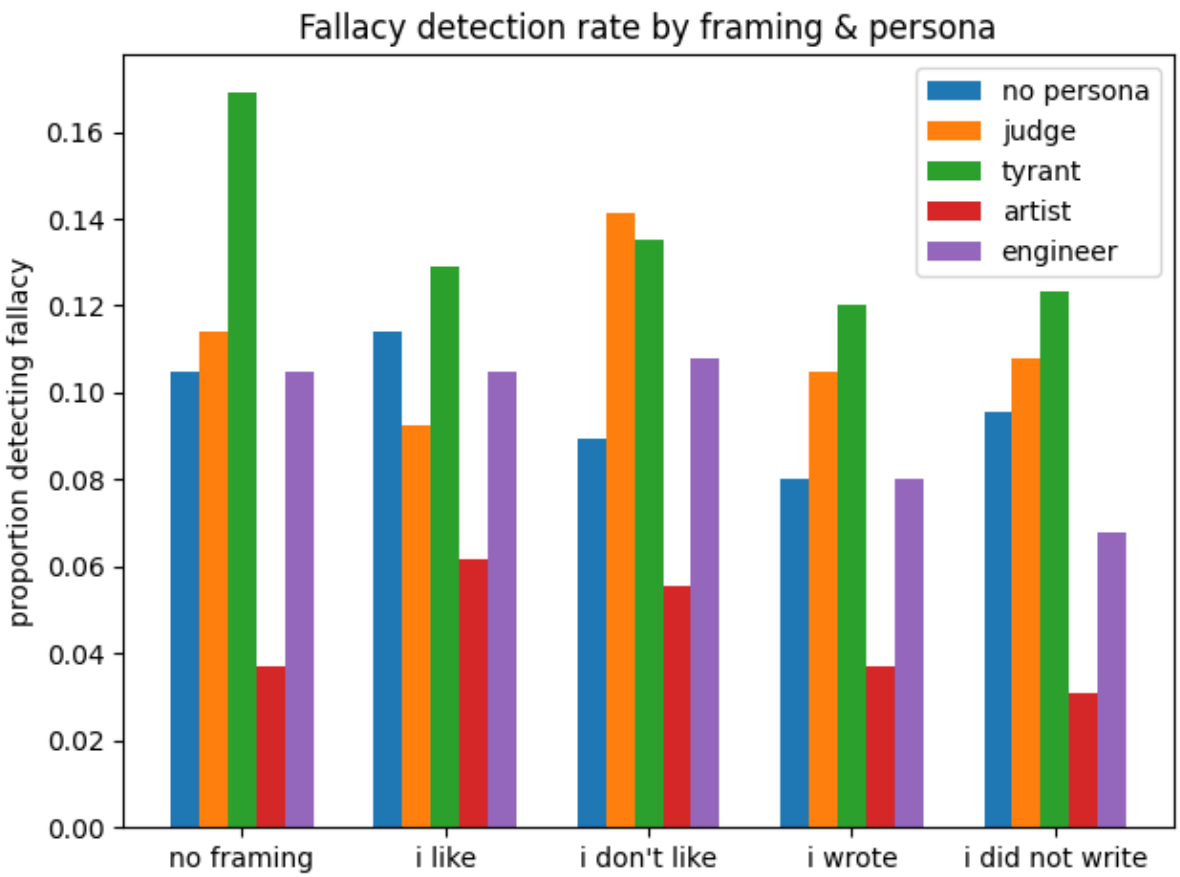

Figure 3: Fallacy detection rate by framing and persona. The Tyrant persona consistently detects fallacies at the highest rate; the Artist at the lowest.

meaningfully attenuate or amplify sycophancy. This asymmetry between the *I wrote it* and *I did not write it* conditions implies that ownership framing operates through a positive attribution mechanism rather than simply modulating perceived stakes.

5.3 Fallacy Detection

Fallacy detection rates showed a different pattern from sentiment and were less variable overall (Figure 3). The Tyrant persona produced the highest detection rates across nearly all framing conditions, reaching approximately 17% under no-framing (well above the no-persona baseline of roughly 10%). The Artist persona consistently showed the lowest detection rates, often falling below 6%, confirming that the Artist's high positive sentiment is accompanied by genuine attenuation of critical evaluation rather than mere tonal softening. Counterintuitively, the Judge persona did not outperform the no-persona baseline, suggesting that simply telling the LLM to be impartial and grounded in its decisions does not reliably induce sharper logical scrutiny.

Framing effects on fallacy detection, shown as lift and drop relative to the no-framing baseline in Figure 4, reveal a consistent sycophantic pattern. Ownership framing (*I wrote it*) produced the largest suppression of fallacy detection across almost all personas, with the Tyrant showing the smallest drop and therefore the greatest resistance to this effect. The *I like it* framing similarly suppressed detection for most personas, while the *I don't like it* framing produced modest improvements in detection for the Judge and Artist, consistent with the interpretation that expressed disapproval partially licenses critical response.

Importantly, fallacy detection and sentiment were partially dissociated: in some conditions the model

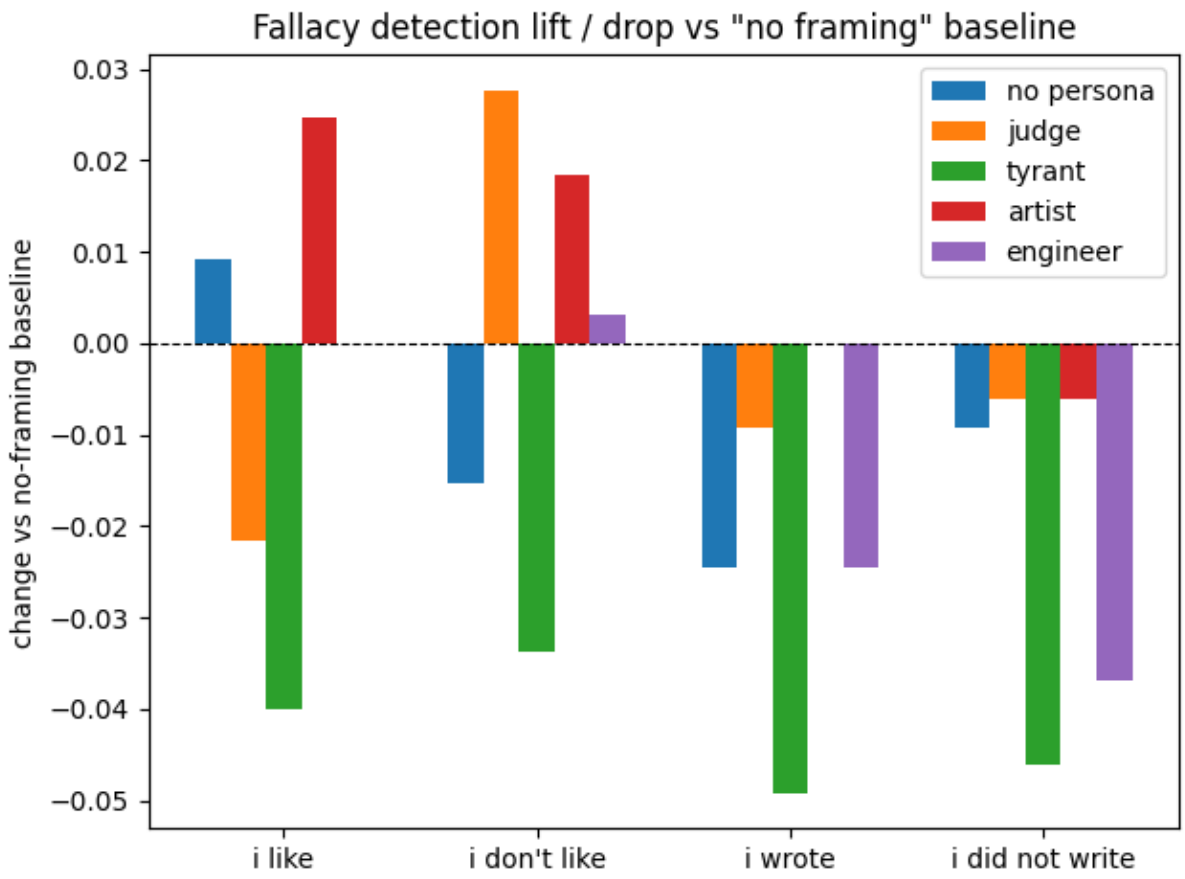

Figure 4: Fallacy detection lift/drop relative to the no-framing baseline. Ownership framing (*I wrote it*) consistently suppresses fallacy detection across personas, with the Tyrant most resistant to this effect.

identified the logical error while still providing an overall positive or validating response. This pattern, which is most visible in the Tyrant and Judge personas under ownership framing, suggests a form of sycophancy in which critique is acknowledged but rhetorically softened, rather than suppressed entirely.

6. Discussion

The central finding of this study is that persona selection exerts a more powerful influence on LLM sycophancy than user driven framing, but I identify several additional patterns.

First, system prompt design is a critical lever for managing sycophancy in deployed systems. Practitioners who wish to minimize sycophantic behavior should pay careful attention to the character and epistemic norms embedded in their system prompts. Some personas may be significantly more susceptible to user preferences at the expense of truth and reality.

Second, and relatedly, the Artist result is striking in its magnitude. A persona that emphasizes beauty, feeling, and openness appears to suppress the model's critical faculty even when presented with objectively flawed arguments. This is not necessarily a design flaw, but it signals that persona selection must be deliberate and context-appropriate.

Third, the Tyrant persona's performance is counterintuitive. Despite being designed to be arrogant and dismissive, the Tyrant was more balanced in its sycophancy than the Judge. I see two possibilities for this. It is certainly possible that the model interpreted the persona of the Tyrant as "unafraid of honesty" rather than "unfiltered critic." It is also possible that the brash attitude of the Tyrant is essentially

counterbalancing the persona that (I assume) user-facing LLMs are deployed with. Because people tend to prefer LLMs that are agreeable, it is not unlikely that companies bias LLMs in that direction, and the Tyrant persona undoes that to the extent that the LLM becomes functionally neutral.

Fourth, the near-equivalence of the no-framing and *I did not write this* conditions is theoretically interesting. It suggests that explicitly distancing oneself from an argument does not prompt more objective feedback from the model. The model behaves as if neutral regardless of authorship disavowal.

Finally, these findings must be interpreted in light of Crawford [2021]'s observations about power and AI. The individuals leading the development and deployment of this technology do so in such a way that their own perspectives are reflected. With that knowledge, it is hard to see the extent to which the Engineer aligns with the "no framing" case and the "no persona" case and not read that as a clear indication that tech companies are writing AI that primarily reflects their homogeneous and narrow views of the world.

7. Conclusions

This paper presents a controlled study of how persona assignment and argument framing independently and interactively shape LLM sycophancy, measured via fallacy detection rate and response sentiment across 25 persona-framing combinations. The central result is clear: who the model is told to be matters more than how the user presents an argument. Between-persona differences in positive sentiment rates spanned nearly 60 percentage points—from the Engineer's near-zero baseline to the Artist's 59%—while within-persona framing effects rarely exceeded 20 percentage points. Ownership framing (*I wrote this argument*) consistently produced the largest within-persona lift, but even this effect was modest relative to the variance explained by persona alone.

Beyond the headline finding, several more granular results deserve emphasis. The Artist persona's extreme sensitivity to framing— swinging 50 percentage points between the *I don't like it* and *I wrote it* conditions—suggests that high-agreement personas are not merely more sycophantic on average but more volatile: more susceptible to amplification by favorable framing and more susceptible to collapse under unfavorable framing. The Tyrant, despite being designed as an adversarial archetype, produced the most consistent fallacy detection rates and the greatest resistance to ownership framing, raising the possibility that bluntness and agreeableness are more orthogonal than they appear. And the dissociation between fallacy detection and sentiment—whereby models sometimes acknowledged a logical flaw while still producing an overall positive response—points to a form of sycophancy that operates at the level of rhetorical framing rather than factual assessment, and that is therefore harder to detect and correct.

Taken together, these results reframe sycophancy as a structural property of the model-persona-framing system rather than a fixed characteristic of any particular model. This has a direct practical implication: system prompt design is not merely a product decision but an epistemic one. The character norms embedded in a system prompt—the values, social orientations, and relational stances ascribed to the model—set the parameters within which user-level framing effects play out. Practitioners deploying LLMs in domains where honest critical engagement matters, from education to medical decision support to legal analysis, should treat persona design with the same rigor applied to output filtering or factual grounding.

More broadly, this study contributes empirical texture to Crawford [2021]'s argument that AI systems embed and reproduce the values of their builders. The finding that the Engineer persona most closely resembles the no-persona baseline is not incidental: it reflects an industry whose defaults are shaped by a particular professional culture. Understanding which personas amplify or attenuate sycophancy is therefore not only a technical question but a political one—about whose epistemic norms get encoded as the model's neutral starting point, and whose critical voice users must explicitly invoke.

7.1 Limitations

Several limitations should be noted. First, the study was conducted with a single LLM; results may not generalize across model families or to models trained with different RLHF objectives. Second, the personas tested were deliberately stylized archetypes; real-world deployments typically use more nuanced and task-specific descriptions. Third, my outcome measures are coarse binary codes; future work should develop more granular metrics for sycophancy. Fourth, the argument stimuli varied in topic and fallacy type, which may introduce uncontrolled variance.

7.2 Future Directions

The most immediate extension of this work is replication across multiple LLM families—GPT, Gemini, Llama, and others—to determine whether the persona-sycophancy relationships observed here are specific to the model tested or reflect more general properties of instruction-tuned language models. Alongside this, future work should move beyond the stylized archetypes used here toward more naturalistic persona descriptions drawn from real deployment system prompts, which tend to be more contextually embedded and professionally specific. A third important direction is the investigation of multi-turn interactions: the present study examines single-turn responses, but sycophancy in practice often compounds across a conversation as the model accumulates social context and the user's expressed preferences become harder to contradict. Future work might also explore deliberate "anti-sycophancy" persona designs. In light of these findings, there are likely personas that encourage the model to prioritize accuracy over agreeableness.

References

Persona-L: LLM-based personas for complex needs in UX design. In *Proceedings of the ACM CHI Conference on Human Factors in Computing Systems*, 2025. URL https://dl.acm.org/doi/10.1145/3706598.3713445.

Myra Cheng, Cinoo Lee, Pranav Khadpe, Sunny Yu, Dyllan Han, and Dan Jurafsky. Sycophantic AI decreases prosocial intentions and promotes dependence. *Science*, 391:eaec8352, 2026. doi: 10.1126/science.aec8352. URL https://www.science.org/doi/10.1126/science.aec8352.

Kate Crawford. *Atlas of AI: Power, Politics, and the Planetary Costs of Artificial Intelligence*. Yale University Press, New Haven, CT, 2021.

Stefano De Paoli. User personas, ideation and large language models: A post-hoc study. *International Journal of Human-Computer Studies*, 208:103690, 2026. doi: 10.1016/j.ijhcs.2025.103690. URL https://www.sciencedirect.com/science/article/pii/S1071581925002472.

Magda Dubois, Cozmin Ududec, Christopher Summerfield, and Lennart Luettgau. Ask don't tell: Reducing sycophancy in large language models, 2026. URL https://arxiv.org/abs/2602.23971. Submitted 27 Feb 2026, last revised 17 Mar 2026.

Aaron Fanous, Jacob Goldberg, Ank A. Agarwal, Joanna Lin, Anson Zhou, Roxana Daneshjou, and Sanmi Koyejo. SycEval: Evaluating LLM sycophancy, 2025. URL https://arxiv.org/abs/2502.08177.

Hugging Face. Distilbert base uncased finetuned sst-2, 2019. URL https://huggingface.co/distilbert/distilbert-base-uncased-finetuned-sst-2-english.

J. M. Korhonen. Notes: Examples of LLMs as synthetic human responders and agents. https://jmkorhonen.wpcomstaging.com/2025/10/29/notes-examples-of-llms-as-synthetic-human-responders-and-agents/, 2025.

Philippe Laban, Lidiya Murakhovs'ka, Caiming Xiong, and Chien-Sheng Wu. Are you sure? Challenging LLMs leads to performance drops in the FlipFlop experiment, 2023. URL https://arxiv.org/abs/2311.08596. Preprint, revised February 2024.

OpenReview. Digital twins of survey respondents via LLM personas. https://openreview.net/forum?id=qh9eGtMG4H, 2024.

Ethan Perez, Sam Ringer, Kamilē Lukošiūtē, Karina Nguyen, Edwin Chen, Scott Heiner, Craig Pettit, Catherine Olsson, Sandipan Kundu, Saurav Kadavath, Andy Jones, Anna Chen, Ben Mann, Brian Israel, Bryan Seethor, Cameron McKinnon, Christopher Olah, Da Yan, Daniela Amodei, Dario Amodei, Dawn Drain, Dustin Li, Eli Tran-Johnson, Guro Khundadze, Jackson Kernion, James Landis, Jamie Kerr, Jared Mueller, Jeeyoon Hyun, Joshua Landau, Kamal Ndousse, Landon Goldberg, Liane Lovitt, Martin Lucas, Michael Sellitto, Miranda Zhang, Neerav Kingsland, Nelson Elhage, Nicholas Joseph, Noemí Mercado, Nova DasSarma, Oliver Rausch, Robin Larson, Sam McCandlish, Scott Johnston, Shauna Kravec, Sheer El Showk, Tamera Lanham, Timothy Telleen-Lawton, Tom Brown, Tom Henighan, Tristan Hume, Yuntao Bai, Zac Hatfield-Dodds, Jack Clark, Samuel R. Bowman, Amanda Askell, Roger Grosse, Danny Hernandez, Deep Ganguli, Evan Hubinger, Nicholas Schiefer, and Jared Kaplan. Discovering language model behaviors with model-written evaluations. 2022. doi: 10.48550/arXiv.2212.09251. URL https://arxiv.org/abs/2212.09251.

Mrinank Sharma, Meg Tong, Tomasz Korbak, David Duvenaud, Amanda Askell, Samuel R. Bowman, Newton Cheng, Esin Durmus, Zac Hatfield-Dodds, Scott R. Johnston, Shauna Kravec, Timothy Maxwell, Sam McCandlish, Kamal Ndousse, Oliver Rausch, Nicholas Schiefer, Da Yan, Miranda Zhang, and Ethan Perez. Towards understanding sycophancy in language models. In *Proceedings of the*

Twelfth International Conference on Learning Representations, 2024. doi: 10.48550/arXiv.2310.13548. URL https://arxiv.org/abs/2310.13548.

Yu-Min Tseng, Yu-Chao Huang, Teng-Yun Hsiao, Wei-Lin Chen, Chao-Wei Huang, Yu Meng, and Yun-Nung Chen. Two tales of persona in LLMs: A survey of role-playing and personalization. In Yaser Al-Onaizan, Mohit Bansal, and Yun-Nung Chen, editors, *Findings of the Association for Computational Linguistics: EMNLP 2024*, pages 16612–16631, Miami, Florida, USA, November 2024. Association for Computational Linguistics. doi: 10.18653/v1/2024.findings-emnlp.969. URL https://aclanthology.org/2024.findings-emnlp.969/.

Miles Turpin, Julian Michael, Ethan Perez, and Samuel R. Bowman. Language models don't always say what they think: Unfaithful explanations in chain-of-thought prompting. *arXiv preprint arXiv:2305.04388*, 2023.

Jason Wei, Xuezhi Wang, Dale Schuurmans, Maarten Bosma, Fei Xia, Ed Chi, and Denny Zhou. Chain-of-thought prompting elicits reasoning in large language models. In *Advances in Neural Information Processing Systems*, volume 35, 2023.